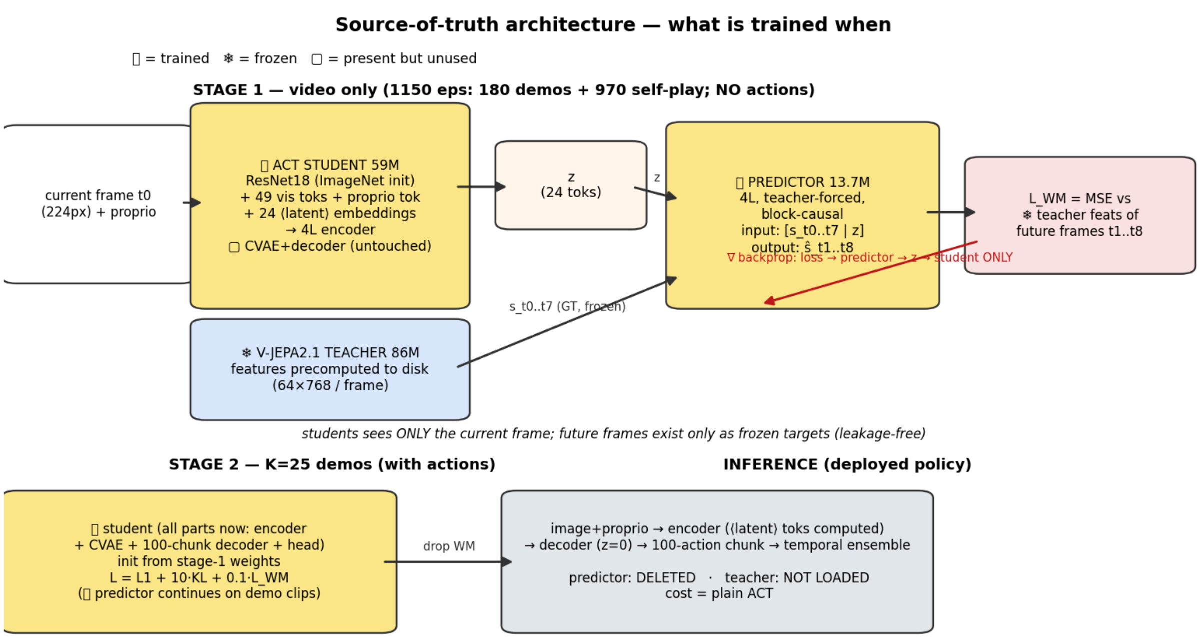
Embodiment via Self-Play–Fed World-Model Teaching

LeRobot research — aloha sim (bimanual insertion) · VLA-JEPA recipe (arXiv 2602.10098) with an ACT student · June 10–12, 2026 · branch `proto/self-play-embodiment`

> **Question.** Can a world-model module, trained on abundant *cheap* same-embodiment data (self-play), teach a small VLA so it needs fewer demos and generalizes better?

> **Headline (3 training seeds × 96 eval seeds).** Feeding **1000 episodes of scripted self-play video** into the world-model corpus makes the K=25-demo policy **+14.3 pts better in-distribution (68.3% vs 54.0% grasp-SR, ≈3.5σ, replicates per-seed)** and ahead at **every** out-of-distribution spawn widening (11/12 cells, sign-test p≈0.003). The same mechanism *without* self-play stays neutral — **the cheap data is the active ingredient**, exactly the original bet.

The validated recipe



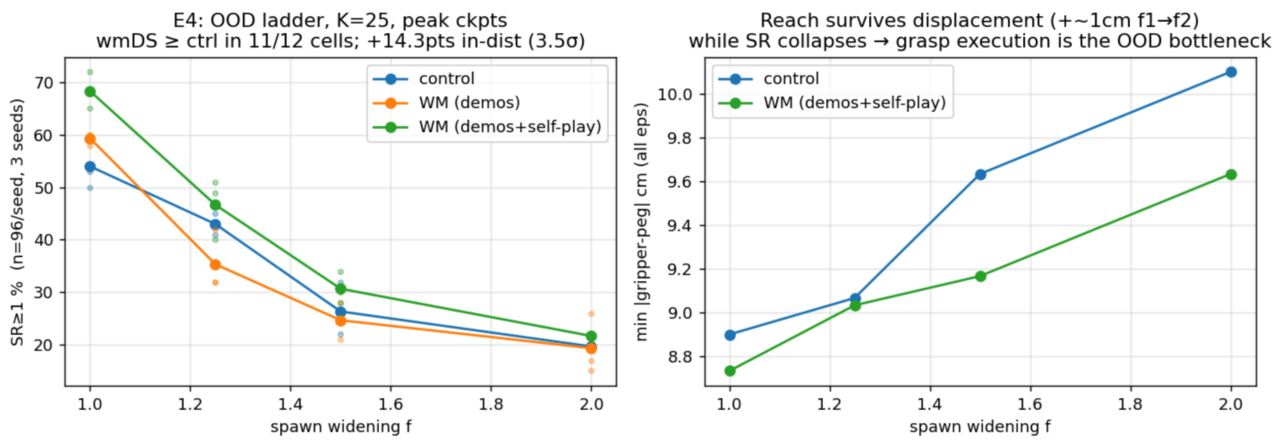
- **Self-play enters as video only** — the WM label is the future itself, so unscripted/imperfect behavior cannot corrupt training (actions never used).
- Faithful VLA-JEPA structure at 59M scale, verified against the paper + the upstream LeRobot port by an **executable self-test gate** (causality probes, leakage tests, mask-equality vs upstream, determinism) — training refuses to start if any check fails.

Result 1 — the module learns transferable anticipation (representation gate)

stage-1 corpus	wide-scene prediction (val/copy ↓)	wide-scene shuffle-z gap ↑
demos only	2.04 (worse than copying)	+1.3% (dead)
demos + self-play	**0.574**	**+36.2%**

On **held-out wide layouts**, the demos-only module mispredicts and its latent-action channel carries nothing. With self-play in the corpus, anticipation transfers (3.5× better prediction; latent tokens carry scene-specific future info). In-distribution performance unchanged — the diversity costs nothing.

Result 2 — it converts to policy gains (3 seeds, n=96, peak protocol)



SR≥1 (grasp), pooled	in-dist	1.25× spawn	1.5×	2×
control (no WM)	54.0%	43.0%	26.3%	19.7%
WM, demos-only corpus	59.3%	35.3%	24.7%	19.3%
WM, demos+self-play	**68.3%**	**46.7%**	**30.7%**	**21.7%**

- Honest framing: the win is a **higher level everywhere** + small absolute OOD edge — *not* a flatter relative decay (our pre-registered slope prediction failed).
- World model is **dropped at inference** — deployment cost identical to plain ACT.

Failure analysis — what still breaks (and what doesn't)

- **Reach generalizes:** grippers get within ~9 cm of displaced objects even at 2× widening (only ~1 cm worse than in-dist), failures included — measured across all 1,152 OOD episodes after first being spotted by eye in replay datasets.
- **Grasp/insert execution is the bottleneck**, in-distribution and OOD alike; secondary mode = late-episode instability past the demo-length tail. ⇒ next gains come from **grasp-diverse demos or grasp-capable play**, not more WM teaching.

What did NOT work (and saved us from scaling blind)

attempt	verdict
Bottleneck embodiment token on frozen encoder (Fork-2)	information-theoretically redundant for a competent policy
Fine-tuning the vision encoder with dynamics objectives (Fork-1/DynaMo)	injects body, **evicts the object**; hurts policy at every demo count
WM teaching with task-redundant corpus (demos only)	neutral-to-harmful; dose-dependent interference at small scale
2B-scale VLA-JEPA reproduction	no-go: 8×A100 recipe; at our scale the in-dist effect (<1%) is below eval noise

Method practices the team can reuse

- **Paper-anchored code + executable self-test gate** — every claim in the paper maps to a runtime assertion; caught 3 silent deviations that had reversed an experimental conclusion.
- **Never compare arms at matched budget** — fixed steps can prove gains but not deficits; use per-arm stopping rules (val-plateau) and peak-over-checkpoints.
- **Closed-loop SR noise:** n=24 swings ±14 pts across eval sets; n≥96 × ≥3 training seeds before believing any gap.
- **Visual failure review** (replayed eval episodes as GUI datasets) re-directed the OOD diagnosis from "reach lock-in" (wrong) to "grasp execution" (measured).

Recommended next steps

- **Option A — scale the validated recipe:** bigger student / real-ALOHA corpus arm / more & richer self-play. Evidence basis now exists.
- **Option B — attack the measured bottleneck:** grasp-diverse demonstrations or grasp-capable play (feeds both representation and action channels).
- All code, raw results, and per-seed data: `research/self_play_embodiment/` (FINDINGS.md, scripts/, results/, figures/), commits `53e9cca2d…10800aec0`.

Generated 2026-06-12 · experiments: aloha sim AlohaInsertion-v0, ACT-style student (59M) + frozen V-JEPA 2.1 teacher · 3 training seeds × 96 paired eval seeds per OOD cell · single RTX 5090